

PYTHON PROJECT DOCUMENTATION

Student Performance Analytics System

Mini Project – CodeAlpha Internship

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Course: Python with AI

1. Project Title

Student Performance Analytics System – a mini project completed during the Python with AI CodeAlpha Internship.

2. Objective

The objective of this project is to analyze student performance using Python, NumPy and Pandas. The project calculates total marks, average marks, grades and pass/fail results. It also performs subject-wise analysis and identifies the top-performing students.

3. Technologies Used

- Python
- NumPy
- Pandas
- Visual Studio Code
- GitHub

4. Dataset Description

The dataset contains 20 student records. Each record contains Student ID, Name, Department, marks in six subjects and Attendance. The six subjects are Python, Maths, English, Physics, Chemistry and Biology. The data is stored in the CSV file students.csv.

Field	Description
Student_ID	Unique student ID
Name	Student name
Department	Student department
Python, Maths, English,Physics,Chemistry,Biology	Marks in the six subjects
Physics, Chemistry, Biology	Marks in the three additional subjects
Attendance	Student attendance percentage/value in the dataset

5. Implementation

1. The CSV dataset is loaded using Pandas with `pd.read_csv()`.
2. The data is organized and displayed as a Pandas DataFrame.
3. The program checks the column names, number of students and basic DataFrame information.
4. Total marks are calculated by adding the marks of all six subjects.
5. Average marks are calculated by dividing total marks by six and rounding the result to two decimal places.
6. The `calculate_grade()` function assigns grades using if-elif-else conditions.
7. A for loop processes each student's average and stores the corresponding grade.
8. The `check_result()` function determines Pass or Fail using an average of 50 as the passing condition.
9. A second for loop processes each student's result.
10. NumPy is used to calculate the highest average, lowest average, class average and subject-wise averages.
11. Pandas sorting is used to identify the top three students by average marks.

Grading Rule:

Average	Grade
90 and above	A+
80–89.99	A
70–79.99	B
60–69.99	C
50–59.99	D
Below 50	F

Pass/Fail Rule: A student is marked Pass when the average is 50 or above; otherwise the result is Fail.

6. Key Features

- Reads student data from a CSV file.
- Uses a Pandas DataFrame for organizing the data.
- Calculates total and average marks for six subjects.
- Assigns grades using a reusable function.
- Determines Pass/Fail using a reusable function.
- Uses loops for processing student averages.
- Uses NumPy for numerical calculations.
- Calculates highest, lowest and class average.
- Performs subject-wise average analysis.
- Identifies the best subject.
- Identifies the top three students.

7. Output / Screenshots

The following screenshots show the actual output of the completed program.

```

Column Names:
Index(['Student_ID', 'Name', 'Department', 'Python', 'Maths', 'English',
      'Physics', 'Chemistry', 'Biology', 'Attendance'],
      dtype='str')

Number of Students:
20

Basic Information:
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Student_ID  20 non-null    str
1   Name        20 non-null    str
2   Department  20 non-null    str
3   Python      20 non-null    int64
4   Maths       20 non-null    int64
5   English     20 non-null    int64
6   Physics     20 non-null    int64
7   Chemistry   20 non-null    int64
8   Biology     20 non-null    int64
9   Attendance  20 non-null    int64
dtypes: int64(7), str(3)

```

Figure 1. Column names, number of students and basic DataFrame information.

```

Student Performance:
  Student_ID  Name Department  Python  Maths  English  ...  Biology  Attendance  Total  Average  Grade  Result
0      S001  Dhiraj      BCA      28     82      75  ...      52           75     353     58.83     D     Pass
1      S002   Aman      BCA      65     71     68  ...      62           60     404     67.33     C     Pass
2      S003  Priya      BCA      91     48     94  ...      79           81     492     82.00     A     Pass
3      S004   Neha      BCA      56     62     40  ...      76           70     312     52.00     D     Pass
4      S005   Rohit      BCA      34     39     81  ...      35           52     278     46.33     F     Fail
5      S006  Anjali      BCA      23     36     30  ...      92           87     256     42.67     F     Fail
6      S007   Karan      BCA      48     55     52  ...      64           72     355     59.17     D     Pass
7      S008   Pooja      BCA      89     22     87  ...      81           79     401     66.83     C     Pass
8      S009   Arjun      BCA      67     64     72  ...      61           57     343     57.17     D     Pass
9      S010  Simran      BCA      47     41     39  ...      30           95     230     38.33     F     Fail
10     S011   Vikas      BCA      39     61     57  ...      69           74     371     61.83     C     Pass
11     S012  Muskan      BCA      93     30     95  ...      85           80     482     80.33     A     Pass
12     S013  Aditya      BCA      71     29     74  ...      51           49     354     59.00     D     Pass
13     S014  Shreya      BCA      42     55     80  ...      33           56     300     50.00     D     Pass
14     S015  Manish      BCA      45     51     49  ...      83           45     381     63.50     C     Pass
15     S016   Nisha      BCA      68     73     66  ...      58           63     407     67.83     C     Pass
16     S017  Deepak      BCA      88     83     85  ...      20           78     444     74.00     B     Pass
17     S018   Komal      BCA      61     58     23  ...      19           84     283     47.17     F     Fail
18     S019   Ravi      BCA      75     70     78  ...      21           48     397     66.17     C     Pass
19     S020   Tanya      BCA      36     89     84  ...      71           68     359     59.83     D     Pass

```

Figure 2. Student Performance table with total, average, grade and result.

```

python -u "d:\Internship Projects\Student_Performance_Analytics\main.py"

Class Analysis:
Highest Average: 82.0
Lowest Average: 38.33
Class Average: 60.02

Subject-wise Average:
Python: 58.3
Maths: 55.95
English: 66.45
Physics: 61.2
Chemistry: 61.1
Biology: 57.1
Best Subject: English

Result:
Passed Students: 16
Failed Students: 4

Top 3 Students:
  Name  Average  Grade
2  Priya   82.00    A
11 Muskan  80.33    A
16 Deepak  74.00    B
PS D:\Internship Projects\Student_Performance_Analytics>

```

Figure 3. Class analysis, subject-wise averages, best subject, pass/fail count and top three students.

8. Final Outcome

The project successfully analyzed the performance of 20 students. The final output showed:

Analysis	Result
Highest Average	82.00
Lowest Average	38.33
Class Average	60.02
Passed Students	16
Failed Students	4
Best Subject	English

Top 3 Students:

Student	Average	Grade
Priya	82.00	A
Muskan	80.33	A
Deepak	74.00	B

9. Challenges & Learning

During the project, I learned how to read CSV data using Pandas and work with DataFrames. I practiced using functions, loops and conditions to process student records. I also learned how to use NumPy for numerical calculations and how to organize a small Python project.

One practical challenge was updating the calculations after using six subjects instead of three. The total and average calculations were updated accordingly, and the final output was tested to check the results.

10. GitHub Repository

Repository: <https://github.com/iamdhiraj-kumar/student-performance-analytics.git>

Repository contents: main.py, students.csv and README.md. The project documentation